

IoT - Intelligent and Connected Systems

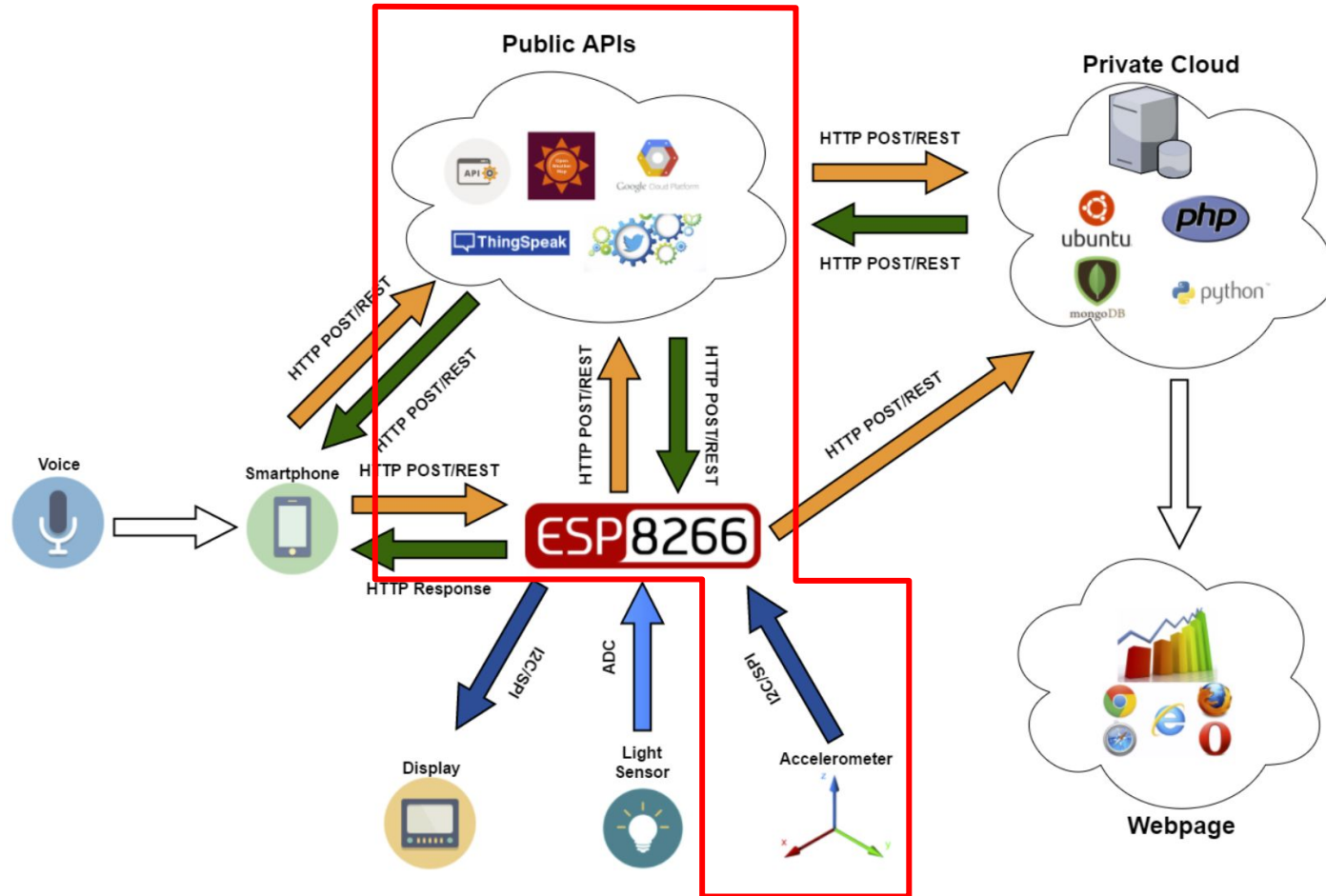
EECS E4764 Fall' 22

Lab 4

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Lab Overview



Part 1

Connecting to the Accelerometer

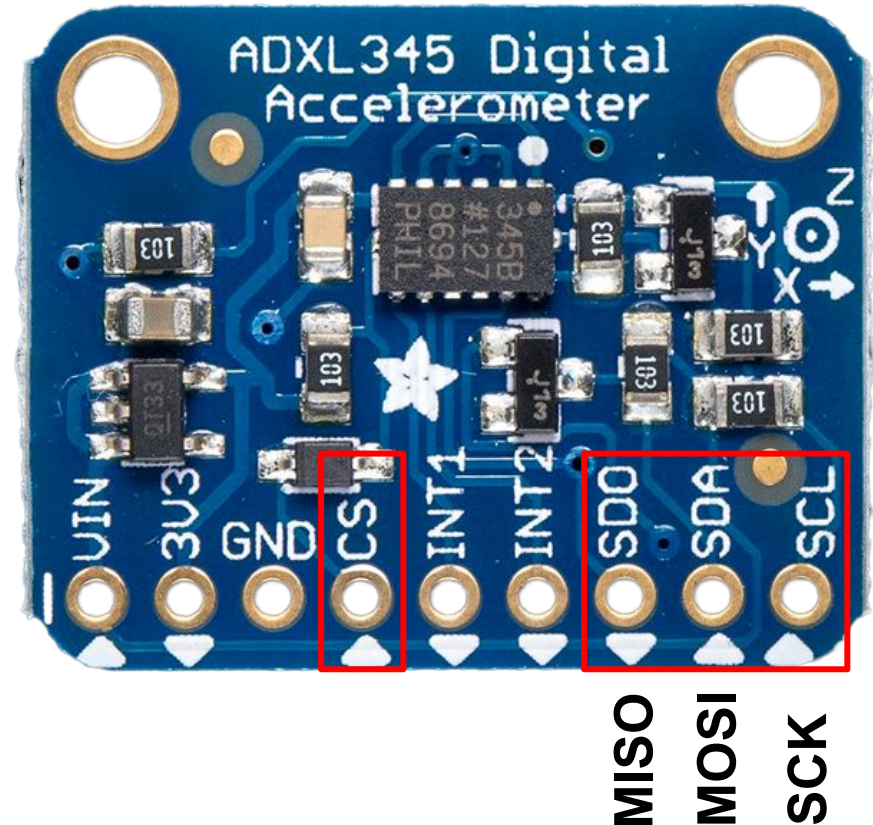
Connecting to Accelerometer (ADXL345)

In this lab, you need to interface with the accelerometer using **SPI**.

The direction of the triangle means input or output. In this lab, VIN, GND, CS, SDO, SDA, SCL need to be connected to ESP.

The datasheet has everything you need:

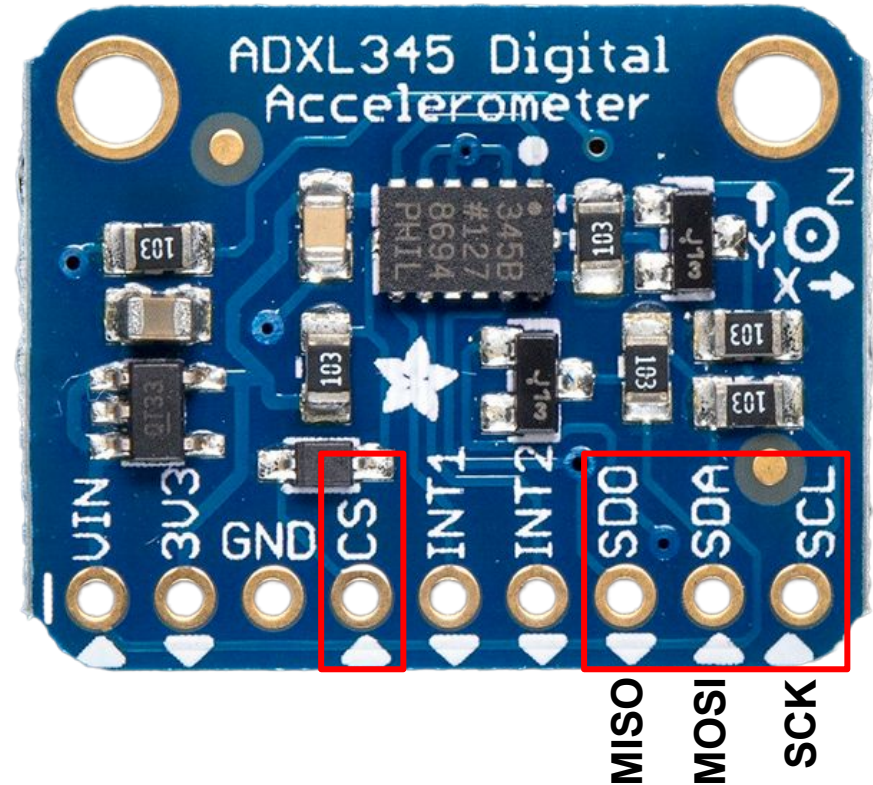
<https://www.sparkfun.com/datasheets/Sensors/Accelerometer/ADXL345.pdf>



Initialization of ADXL345

Before you initialize the sensor, you need to know how to write data to the sensor. Details in the datasheet page 9 (SPI 4-wire)

The registers you need to initialize are **0x31**, **0x2d**, **0x2c**, **0x2e**, **0x38**. Please read the register map at page 14 and find out what value you should write in.



Sample code for spi writing

```
1  import machine
2  import time
3
4  # initial hardware spi on huzzah
5  hspi = machine.SPI(1, baudrate=1500000, polarity=1, phase=1)
6  cs = machine.Pin(15, machine.Pin.OUT)
7  # enable adxl345 by setting chip select signal low
8  cs.value(1)
9  time.sleep_ms(50)
10 cs.value(0)
11 # write data
12 hspi.write(b'\x21\x00')
```

Reading from ADXL345

Details in datasheet page 9.

Be careful that the first bit (read/write) need to be set to **1**, in order to enter reading mode.

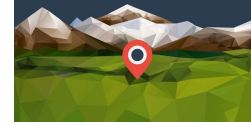
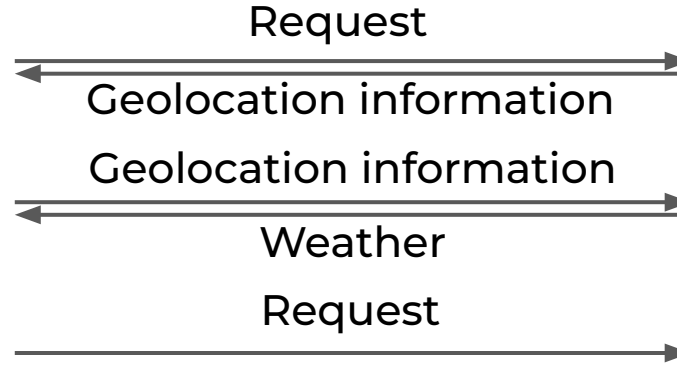
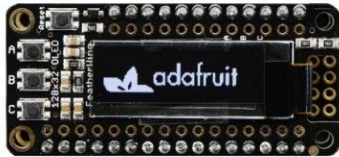
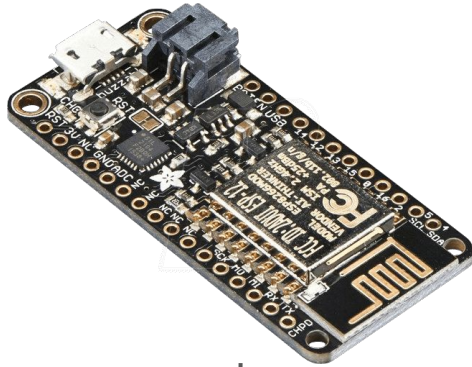
You can use **SPI.readinto** function

<https://docs.micropython.org/en/latest/library/machine.SPI.html#machine.SPI.readinto>

Part 2

Connecting to the Internet and using APIs

Overview



What do you need to do?

- Connect the Huzzah to the Internet. You will need to know the WiFi name and password to access the network.
- Once you are connected to the internet, you need to make a REST call to fetch your location information. Feel free to use any of the available APIs which can give you this information. Store the location information for future reference in your code. Print your latitude and longitude on the OLED.
- Use your location information to get the weather information from another API. Print your Weather information on the OLED. OLED has limited display space, so printing the temperature in your location with a one word weather summary will work. For example “25 C, Sunny”.
- Post your weather to Twitter.

Note

1. Location information means Latitude and Longitude in the context of this lab exercise.
1. There are many other APIs that allow you to send twitter messages, other than the Twitter API; you are welcome to use any of them. The tweets you send do not have to be typed out and can be preset.
1. The same applies to Geolocation and Weather APIs. Feel free to use any API that you are comfortable using.
1. Remember that some APIs might need you to register and/or enter a billing information prior to usage. In most cases, pricing is based on number of concurrent calls and API usage on a monthly basis. Do read the terms carefully before signing up for an API where you have to enter billing information.

Lab 4 Checkpoints

1. On top of the combined system from the last lab, implement text scrolling based on accelerometer readings.
2. Interface with a Geolocation API and display your coordinates on the OLED display.
3. Feed your coordinates into an API to receive information about the weather in your area; display the results on your OLED display.
4. Post a tweet from the ESP8266. An example tweet could be your location coordinates with the weather in that location. Also, open Twitter website or app with your PC or phone to show the tweets.